

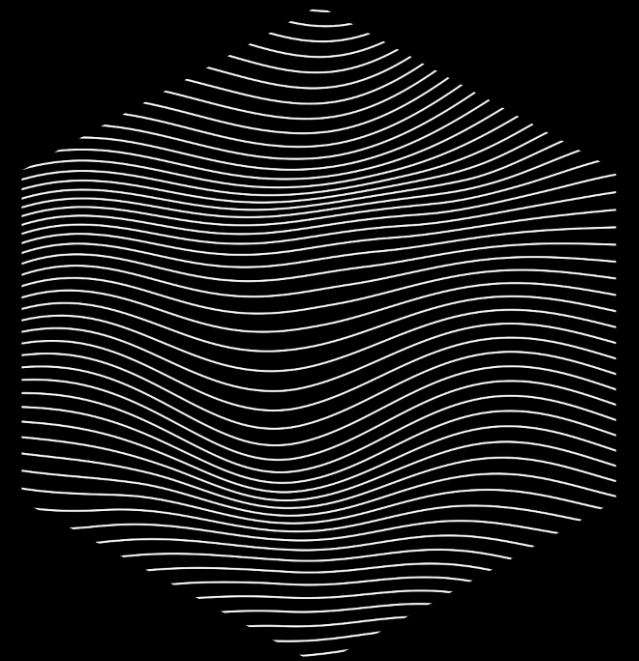


wingfoil

**the ultra low latency data streaming
framework**

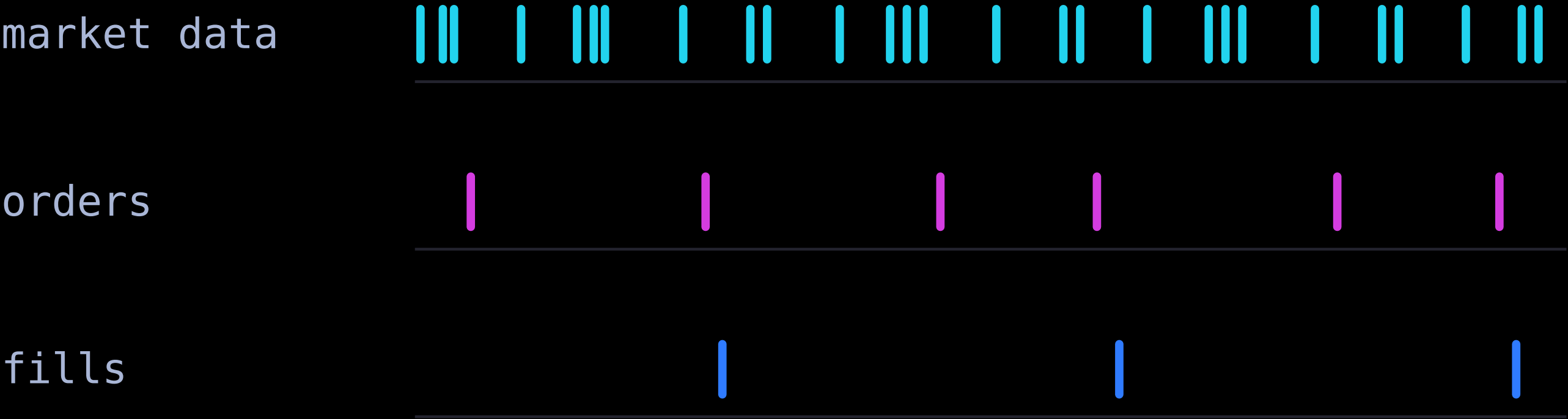
Jake Mitchell

August 2026



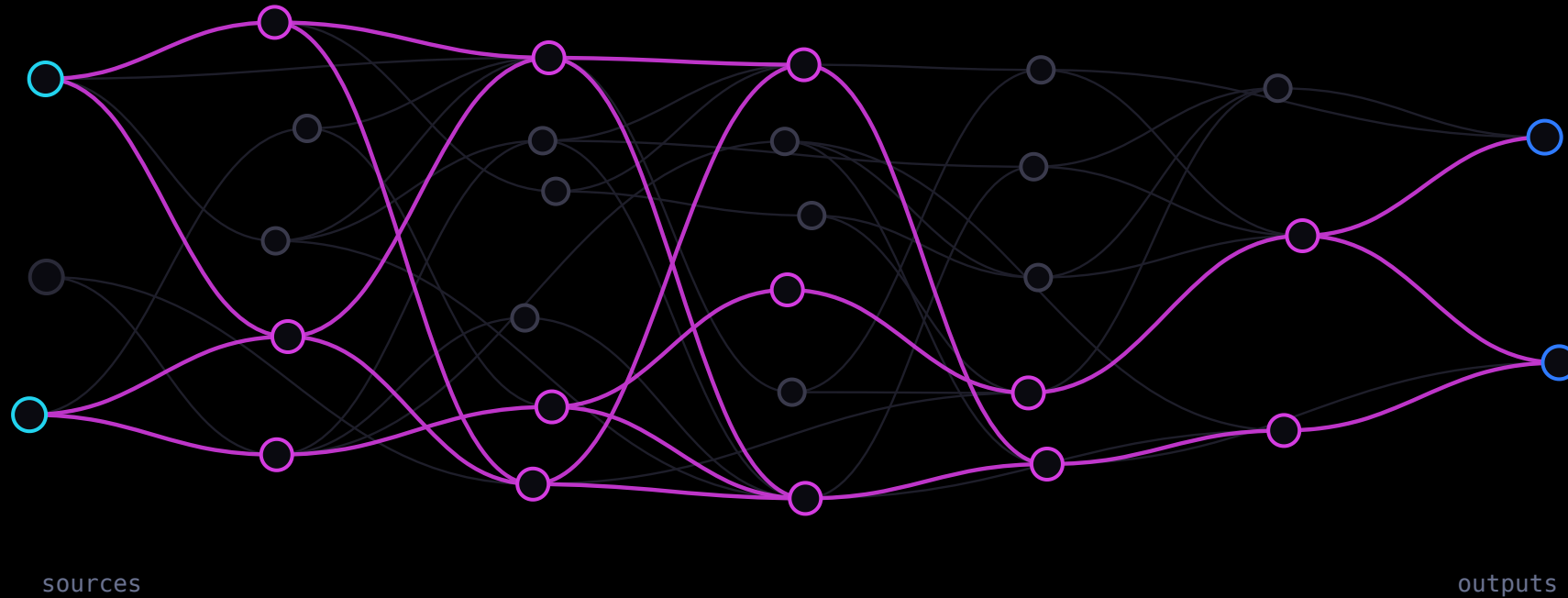
THE PROBLEM

Everything ticks at a different rate



THE SOLUTION

Wire a graph of calculations



- The framework abstracts over **what needs to tick, and when**.
- **Split and recombine** freely — it stays one graph.
- **Modulate the frequency** — window, throttle, sample.
- And it has to be **efficient**.

Wire it, build it, run it

```
let g = GraphBuilder::new();

let printed = g
    .ticker(Duration::from_millis(100))
    .count()
    .map(|i| format!("tick {i}"))
    .for_each(|msg: &String| { println!("{}", msg); Ok(()) });

g.build().run(RunMode::HistoricalFrom(NanoTime::ZERO), RunFor::Cycles(5));
```

```
historical run (instant):
```

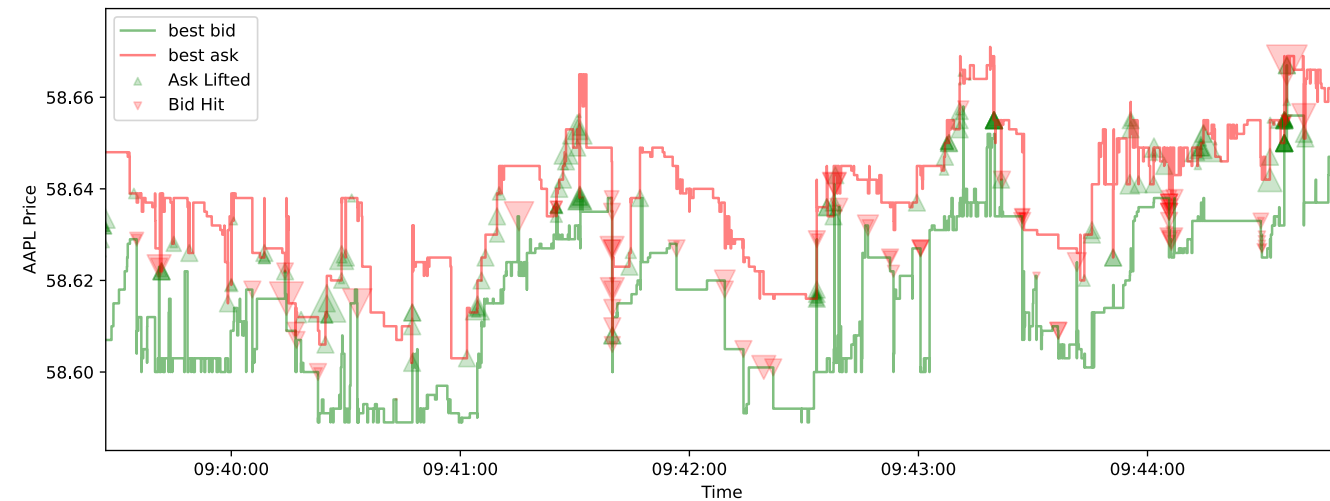
```
tick 1
tick 2
tick 3
tick 4
tick 5
```

```
cargo run -p wingfoil --example hello_graph
```

MOTIVATION, NOT THESIS

A real graph: NASDAQ limit orders → book → prices + fills

```
let (fills, prices) =  
  csv_read(&g, &src, get_time, true, None)?  
    .map(move |chunk: &Burst<Message>| {  
      process_orders(chunk, &book)  
    })  
    .split();  
  
prices.filter_none().distinct()  
  .csv_write(&prices_path)?;  
fills.csv_write(&fills_path)?;
```



RUN MODES

Backtest it, then run it live

● ● ● **RunMode::HistoricalFrom(NanoTime::ZERO)** — the whole hour

```
$ cargo run --release --example top_of_book
0.021_311 bid 585.33 ask 585.91 spread 0.58 mid 585.62
0.197_502 bid 585.33 ask 585.92 spread 0.59 mid 585.62
0.197_540 bid 585.33 ask 585.93 spread 0.60 mid 585.63
0.201_332 bid 585.36 ask 585.93 spread 0.57 mid 585.64
0.267_498 bid 585.73 ask 585.74 spread 0.01 mid 585.74
```

● ● ● **RunMode::RealTime** — a three-second window of the same file · animation slowed 12×

```
$ cargo run --release --example top_of_book -- realtime
1,787,777,083.871_535 bid 585.33 ask 585.91 spread 0.58 mid 585.62
1,787,777,084.048_929 bid 585.33 ask 585.92 spread 0.59 mid 585.62
1,787,777,084.049_141 bid 585.33 ask 585.93 spread 0.60 mid 585.63
1,787,777,084.053_476 bid 585.36 ask 585.93 spread 0.57 mid 585.64
1,787,777,084.119_906 bid 585.73 ask 585.74 spread 0.01 mid 585.74
```

I/O

Sixteen adapters, three patterns

MESSAGING

Kafka · ZeroMQ · Fluvio · Redis · etcd

LOW-LATENCY TRANSPORT

Aeron · iceoryx2

TRADING

FIX

STORAGE & FILES

kdb+ · PostgreSQL · CSV · lines

WEB

WebSocket client · WebSocket server

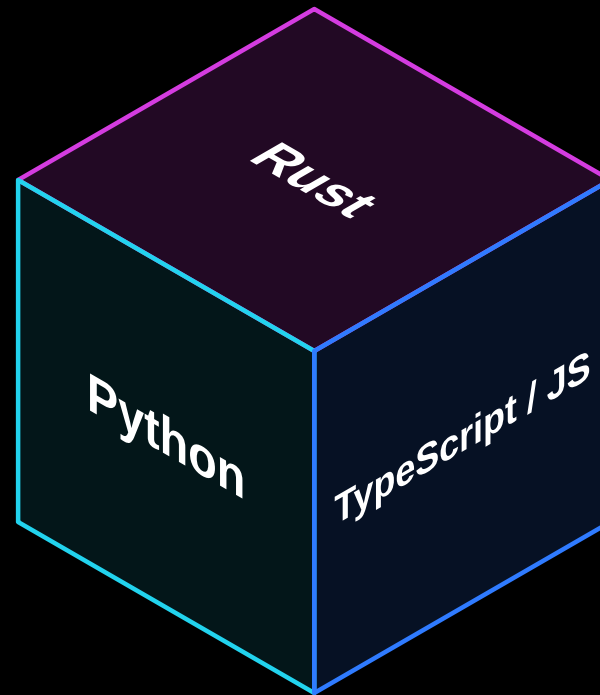
TELEMETRY

Prometheus · OpenTelemetry

Pattern	Latency	Example
Busy loop	nanoseconds	iceoryx2
Worker thread	microseconds	ZeroMQ
Async	microseconds	kdb+

SURFACES

Three languages, one engine



PART TWO

Nitro

Performance, the wall we hit chasing it,
and the pattern that got us through.

THE ENGINEERING PROBLEM

What we wanted: compile the graph away

Interpreted, every node pays a **dynamic dispatch** and a **trip through a value slot** — per node, per cycle. On a hot chain that dominates the arithmetic you came to do.

So: **generate code**. One straight-line function, state in locals, the optimiser working across node boundaries.

We built exactly that — ran the wiring from `build.rs`, walked the graph, emitted a runner. It failed three ways.

- 1 It had to **re-implement every node**.
Semantics were welded to `RefCell` fields, so generated code could not *call* a node's cycle.
- 2 **Types and closures were already gone**.
You cannot get `|x| x * 10` back out of a `Box<dyn Fn>`.
- 3 **Scheduling was inferred from names** — the engine drove nodes off allowlists.

The problem was never the generator. Node semantics were not *separately callable*.

THE FRONT DOOR

You write the wiring once, in plain Rust

```
wingfoil::nitro! {  
    fn odds_evens(g: &GraphBuilder) -> Stream<String> {  
        let count      = g.ticker(PERIOD).count();  
        let is_even    = count.map(|i| i.is_multiple_of(2));  
        let is_odd     = is_even.map(|b| !b);  
        let odd_str    = count.filter(&is_odd).map(|i| format!("{i} is odd"));  
        let even_str   = count.filter(&is_even).map(|i| format!("{i} is even"));  
        let labelled   = odd_str.merge(&even_str);  
        labelled  
    }  
}
```

Out comes `odds_evens::interpreted()`, `odds_evens::compiled()` and `odds_evens::nested()` — from the same tokens.

The wiring function stays **valid, reviewable Rust** either way. That property was deliberately never traded away: `count` is read three times, so it is a shared apex node — the macro derives a real DAG, not a chain.

...AND THIS IS WHAT IT BECOMES

One function. One local per node. No dyn.

```
let mut __k = Kernel::new(run_mode, run_for);

let mut __cfg_wf_anon_1    = PERIOD;
let mut __state_wf_anon_1 = __wf_op_ticker_seed_state(&__cfg_wf_anon_1);
let mut __v_wf_anon_1     = __wf_op_ticker_seed_value(&__cfg_wf_anon_1);
// ... one cfg / state / value local per node, 11 nodes ...

let mut __dirty = [false; 11usize];
while __k.begin_cycle(&mut __dirty) {
    let __t_wf_anon_1 = /* activation */ && {
        let mut __ctx = Ctx::new(&mut __k, 0usize);
        match __wf_op_ticker_cycle(&mut __cfg_wf_anon_1,
                                   &mut __state_wf_anon_1, (), &mut __ctx)
            .context("node 0 (wf_anon_1: ticker) cycle")? {
            Tick::Value(__val) => { __v_wf_anon_1 = __val; true }
            Tick::Silent(__val) => { __v_wf_anon_1 = __val; false }
            Tick::Quiet       => false,
        }
    };
};
// ... 10 more, emitted in topological order ...
}
```

abridged · elisions marked · all 1,730 lines committed verbatim beside the example:
github.com/wingfoil-io/wingfoil → [examples/core/dual_mode/expanded/](https://github.com/wingfoil-io/wingfoil/blob/main/examples/core/dual_mode/expanded/)

THE ONE DECISION EVERYTHING FOLLOWS FROM

Semantics became an associated function

```
trait Op {  
    type Cfg;           // construction-time config; closures live here  
    type State;         // engine-owned mutable state  
    type In<'a>;        // typed inputs, passed in per cycle  
    type Out;           // the produced value  
    const ACTIVATION: Activation;  
  
    fn cycle(cfg: &mut Self::Cfg, state: &mut Self::State,  
            input: Self::In<'_>, ctx: &mut Ctx<'_>) -> Result<Tick<Self::Out>>;  
}
```

Read what is *absent*. Nothing here says how state is stored, where inputs come from, or who calls it.

THE SAME NODE, BOTH WAYS

The node changed completely. The wiring barely moved.

legacy — the node owns everything

```
#[node(active = [upstream], output = value: f64)]
impl MutableNode for ScaleStream {
    fn cycle(&mut self, _state: &mut GraphState)
        -> Result<bool>
    {
        self.value =
            self.upstream.peek_value() * self.factor;
        Ok(true)
    }
}
```

owns: its upstream handles, its state, its output slot

```
let count = ticker(Duration::from_millis(10))
    .count();
count.run(RunMode::RealTime, RunFor::Cycles(100))?;
```

now — the engine owns everything

```
#[op(build = scale, fluent)]
impl Op for Scale {
    type Cfg    = f64;           // config
    type State  = ();           // engine-owned
    type In<'a> = (&'a f64,);    // passed in
    type Out    = f64;
    const ACTIVATION: Activation = Activation::NONE;

    fn cycle(cfg: &mut f64, _state: &mut (),
        input: (&f64,), _ctx: &mut Ctx<'_>)
        -> Result<Tick<f64>>
    {
        Ok(Tick::Value(input.0 * *cfg))
    }
}
```

owns: nothing — cfg, state and inputs all arrive as arguments

```
let g = GraphBuilder::new();
let count = g.ticker(Duration::from_millis(10))
    .count();
g.build().run(RunMode::RealTime, RunFor::Cycles(100))?;
```

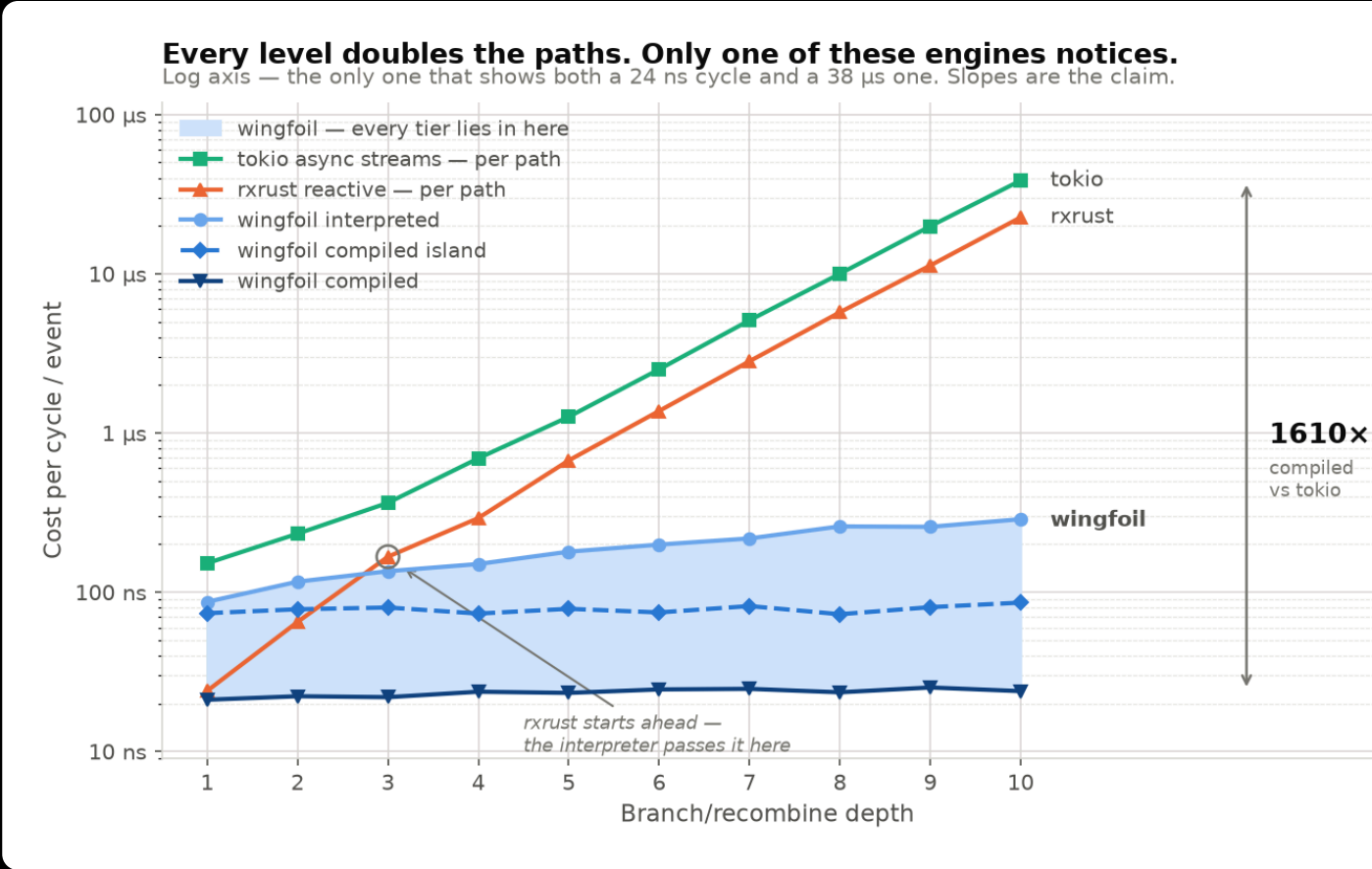
RESULTS

The 10×10 graph: 16.6 ns → 0.31 ns per node cycle

10×10 fan-out	per node cycle		
legacy engine	16.55 ns		
interpreted	11.64 ns	1.4×	
compiled	0.31 ns	53×	
nested island	1.39 ns	12×	

103 nodes, every one ticking, 10 000 cycles — all four measured back to back in one run.

And the shape that separates execution models: branch and recombine, where **paths double every level**. Rx and async streams pay per path — measured slopes **2.01×** and **1.94×**. Every wingfoil tier stays flat.



PART THREE

What's next

Down the stack, up the stack,
and where we could use help.

DOWN THE STACK

Closer to the metal

- 1 **Core pin** — the graph thread on an isolated core.
- 2 **Kernel bypass** — Onload, then ef_vi/DPDK. This is the one that buys the wire-to-trade number we currently decline to claim.
- 3 **Zero-allocation I/O** — no allocator on the hot path.
- 4 **Verilog / FPGA** — the graph as gateware, with the backtest as its testbench.

Same graph definition. Further down.

UP THE STACK

A platform, not just an engine

The missing layer

Simulated exchange · OMS / EMS · position keeping · risk · trade booking · venue adapters

None of it is a new kind of thing

Position keeping is a *fold*. Risk is a *filter*. PnL is a *join*.
The simulated venue is just an *op*.

HELP WANTED

We'd love contributors

- Around **20 people** so far.
- **Good first issues** that really are — each names the file to change and code to copy.
- The bigger projects are separable, and each links to a design rather than a blank page.

Also genuinely useful

Where the docs lost you. The benchmark you don't believe. A project wingfoil would suit — or plainly wouldn't.

THANK YOU



If you work in financial services — or want to build a system to trade your own account — **please do reach out.**

github.com/wingfoil-io/wingfoil · Apache-2.0

```
cargo add wingfoil · pip install wingfoil
```

hello@wingfoil.io · [Discord](#) · [GitHub Discussions](#)

